

Breaking the Combinatorial Explosion: GNN-Guided Pruning for Optimal Bundle Pricing

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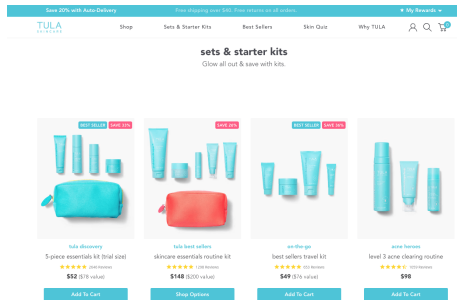
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Bundle pricing is a menu-design problem

BUSINESS SETTING

- ▶ Think of a skincare brand selling cleanser, toner, serum, and moisturizer.
- ▶ It can sell each product separately, or package several products into a starter kit.
- ▶ Customers compare “what the set is worth to me” with the posted price, then pick the most attractive option.
- ▶ Mixed bundling: which individual products and bundles should appear in the menu, and at what prices?



Skincare sets give a concrete menu: individual products coexist with curated bundles and discounted kits.

The MB formulation keeps the economic structure of the problem

FORMULATION For a retained candidate family $\mathfrak{B} \subseteq 2^{[n]}$, let $\mathcal{I}_{\mathfrak{B}}$ index the retained bundles. **Variables:** θ_{kb} indicates which bundle segment k chooses; p_b is the posted price; P_{kb} is the paid price linked to that choice; s_k is surplus; Z_{kb} is profit.

$$\max \sum_{k=1}^m \sum_{b \in \mathcal{I}_{\mathfrak{B}}} \alpha_k Z_{kb}$$

Choice and surplus

$$\sum_{b \in \mathcal{I}_{\mathfrak{B}}} \theta_{kb} = 1, \quad s_k \geq R_{kb} - p_b$$

$$s_k = \sum_{b \in \mathcal{I}_{\mathfrak{B}}} (R_{kb} \theta_{kb} - P_{kb})$$

Each segment chooses one option with maximum surplus.

Price paid and profit

$$p_b - R_{\max}(1 - \theta_{kb}) \leq P_{kb} \leq p_b$$
$$Z_{kb} = P_{kb} - c_{kb} \theta_{kb}$$

Self-selection across segments

$$s_k \geq \sum_{b \in \mathcal{I}_{\mathfrak{B}}} (R_{kb} \theta_{\gamma b} - P_{\gamma b}), \quad \forall k, \gamma.$$

Segment k must not prefer the bundle-price pair chosen by segment γ .

Price subadditivity

$$p_b \leq \sum_{c \in \mathcal{C}} p_c \quad \text{if} \quad A(b) \subseteq \bigcup_{c \in \mathcal{C}} A(c).$$

The restricted model $HM(\mathfrak{B})$ is the same MB logic on a smaller menu.

Exact mixed bundling captures self-selection, but the bundle space explodes

COMPUTATIONAL BOTTLENECK

Combinatorial menu

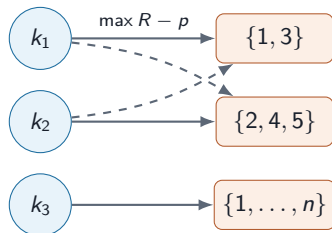
n products $\Rightarrow |\mathcal{F}| = 2^n$ candidate bundles.

Assignment variables: $O(m2^n)$.

Non-additive valuations

$$R_{kb} = f \left(\sum_{j \in A(b)} u_{kj} \right)$$

Concavity captures diminishing marginal utility; non-additivity prevents reducing everything to item-level prices.



- ▶ One price change can move several segments to different bundles.
- ▶ Prices are shared across segments, so choices are strategically coupled.
- ▶ Full MB is accurate, but quickly becomes intractable when n grows.

The paper is about learning a compact optimization representation

QUESTIONS AND CONTRIBUTIONS

Research questions

1. How can we represent non-additive mixed bundling without building the exponential MILP graph?
2. How should GNN predictions be converted into inference policies that effectively prune the intractable 2^n search space into a manageable subset?
3. Can a model trained on exact instances guide larger instances where exact MB is unavailable?

Contributions

- ▶ Compact customer-product graph and edge-output GNN.
- ▶ FCP, PCP-cp, GNN-guided local search and self-improvement.
- ▶ WL-based expressiveness result plus numerical evidence.



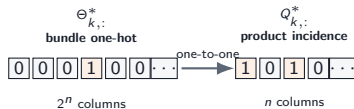
The key trick is to learn an $m \times n$ projection instead of $m \times 2^n$

REPRESENTATION INSIGHT

Do not learn this

$$\Theta^* = (\theta_{kb}^*) \in \{0, 1\}^{m \times 2^n}$$

- ▶ Bundle-level assignment is exponential.
- ▶ Directly predicting a bundle index scales poorly.



Learn this projection

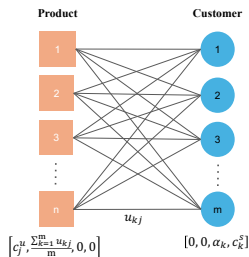
$$q_{kj}^* = \sum_{b:j \in A(b)} \theta_{kb}^*$$

$$Q^* \in \{0, 1\}^{m \times n}.$$

- ▶ $q_{kj}^* = 1$ means product j is in the bundle purchased by segment k .
- ▶ This is exactly an edge-level target on the customer-product graph.

The GNN is built on a customer-product graph and outputs edge probabilities

GNN CONSTRUCTION



- ▶ Product node j : unit cost and average utility

$$y_j^P = \left[c_j^u, \frac{1}{m} \sum_k u_{kj}, 0, 0 \right].$$

- ▶ Segment node k : segment weight and service cost

$$y_k^S = [0, 0, \alpha_k, c_k^s].$$

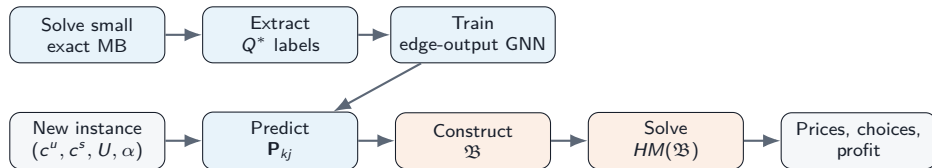
- ▶ Edge (k, j) : segment-product utility u_{kj} .
- ▶ Message passing updates node embeddings and edge embeddings; final edge logits become

$$\mathbf{P} = \sigma(Z^{(\Omega)})^\top \in [0, 1]^{m \times n}.$$

Permutation equivariance; weight sharing support transfer across different m, n .

Learning guides the search space; optimization still solves pricing

FRAMEWORK



What ML does

Predicts product inclusion probabilities on $m \times n$ customer-product edges.

What optimization does

Computes final prices and self-selection assignments on the retained bundle family.

FCP and PCP-cp turn probabilities into candidate bundle families

PRUNING POLICIES

FCP: fast, aggressive

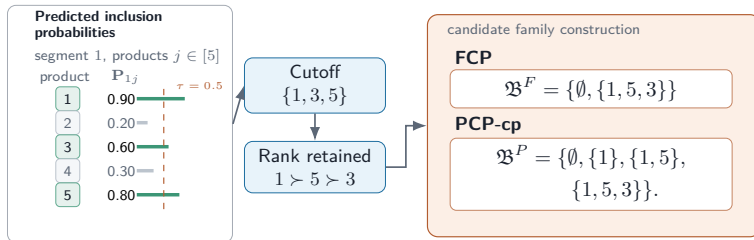
$$\mathfrak{B}^F = \{\emptyset\} \cup \{B_k^F\}_{k=1}^m, \quad B_k^F = \{j : \mathbf{P}_{kj} \geq \tau\}.$$

One predicted bundle per segment.

PCP-cp: richer, conservative

$$\mathfrak{B}^P = \{\emptyset\} \cup \left\{ \{j_{k,(1)}, \dots, j_{k,(i)}\} : i \leq |U_k|, k \in [m] \right\}.$$

Nested prefix bundles after sorting by $\mathbf{P}_{kj}$.



GNN-guided local search and self-improvement

REFINEMENT

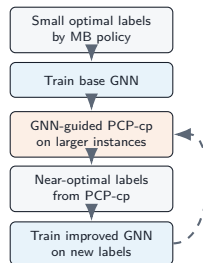
FCP-Local Search

Start from FCP, then evaluate high-confidence add/drop moves:

$$\text{Score}_{kj} = \begin{cases} \mathbf{P}_{kj}, & j \text{ currently unselected,} \\ 1 - \mathbf{P}_{kj}, & j \text{ currently selected.} \end{cases}$$

- ▶ Keep only top- K add/drop candidates globally.
- ▶ Accept first improving move; LP relaxation acts as a fast proxy.

Iterative self-improvement



- ▶ Exact large labels are infeasible.
- ▶ PCP-cp solutions become the next round's training labels.

Customer-Product WL Test with Edge Colors

THEORY I: WL AND UNFOLDABILITY

Algorithm 1 Customer-Product WL Test with Edge Colors, denoted by WL_{CP}

Input: A customer-product graph instance $x = (G, H, E) \in \mathcal{X}_{m,n}$ and the number of WL refinement rounds T .

Initialize:

$$C_k^{0,S} = \text{HASH}_{0,S}(y_k^S), \quad C_j^{0,P} = \text{HASH}_{0,P}(y_j^P), \quad C_{kj}^{0,E} = \text{HASH}_{0,E}(C_k^{0,S}, C_j^{0,P}, e_{kj}).$$

for $\ell = 1, 2, \dots, T$ **do**

$$C_k^{\ell,S} = \text{HASH}_{\ell,S}\left(C_k^{\ell-1,S}, \sum_{j=1}^n e_{kj} \text{HASH}'_{\ell,P}(C_j^{\ell-1,P})\right), \quad \forall k \in \{1, \dots, m\}.$$

$$C_j^{\ell,P} = \text{HASH}_{\ell,P}\left(C_j^{\ell-1,P}, \sum_{k=1}^m e_{kj} \text{HASH}'_{\ell,S}(C_k^{\ell-1,S})\right), \quad \forall j \in \{1, \dots, n\}.$$

$$C_{kj}^{\ell,E} = \text{HASH}_{\ell,E}(C_k^{\ell-1,E}, C_k^{\ell,S}, C_j^{\ell,P}), \quad \forall k \in \{1, \dots, m\}, j \in \{1, \dots, n\}.$$

end for

Return:

$$\left(\{C_k^{T,S}\}_{k=1}^m, \{C_j^{T,P}\}_{j=1}^n, \{C_{kj}^{T,E}\}_{k=1,j=1}^{m,n} \right).$$

Foldable and unfoldable instances

An instance is *foldable* if two distinct customer nodes or two distinct product nodes receive identical node-side WL colors at every refinement level. Otherwise it is *unfoldable*. For an unfoldable instance, node-side colors become discrete after at most $m + n$ rounds, so distinct edge coordinates are also distinguishable after finitely many rounds.

The theorem justifies the compact product-assignment target

THEORY II: REPRESENTATION RESULT

Canonical target

For an optimal MB solution W , define the product-assignment projection

$$Q(W) = \Theta(W)M \in \{0, 1\}^{m \times n}.$$

Because full MB may have multiple optima, define a canonical equivariant mapping Φ_Q by WL-based sorting and lexicographic tie-breaking.

Main result

For any finite solvable non-foldable dataset D , any $\eta \in (0, \frac{1}{2})$, and any $\delta > 0$, there exists an edge-output GNN F_δ such that

$$\|F_\delta(x) - \Psi_Q^{(\eta)}(x)\|_\infty < \delta, \quad \forall x \in D,$$

where $\Psi_Q^{(\eta)} = \text{logit}(\eta \mathbf{1} + (1 - 2\eta)\Phi_Q)$.

Operational corollary

After sigmoid, if the error is below $1/2 - \eta$, thresholding at 0.5 recovers $\Phi_Q(x)$. Thus theory supports the edge-level representation.

Small instances show near-optimal profit with much less computation

RESULTS I: AGAINST EXACT MB

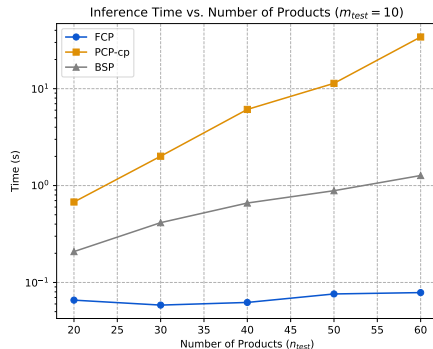
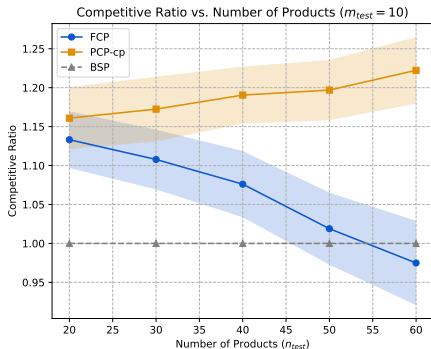
Scenario	FCP			PCP-cp			FCP-Local Search			BSP		
	$CR_{.,MB}$ (std)	Time (s)	$TR_{.,MB}$	$CR_{.,MB}$ (std)	Time (s)	$TR_{.,MB}$	$CR_{.,MB}$ (std)	Time (s)	$TR_{.,MB}$	$CR_{.,MB}$ (std)	Time (s)	$TR_{.,MB}$
$m = 10, n = 10$	0.974 (0.019)	0.056	0.008	0.986 (0.010)	0.292	0.042	0.985 (0.018)	0.212	0.031	0.870 (0.037)	0.058	0.009
$m = 30, n = 10$	0.964 (0.025)	1.703	0.009	0.985 (0.009)	23.373	0.083	0.971 (0.026)	9.341	0.058	0.860 (0.028)	1.356	0.009

Subset of the paper's $n = 10$ comparison table; $CR_{.,MB}$ and $TR_{.,MB}$ are relative to full mixed bundling.

- ▶ All GNN-guided policies recover more than 96% of exact MB profit.
- ▶ PCP-cp is the highest-quality pruning rule on small exact instances.
- ▶ BSP is fast, but loses around 13%–14% profit.
- ▶ The result validates pruning quality before moving to larger n .

Large instances reveal a quality-speed frontier

RESULTS II: SCALABILITY



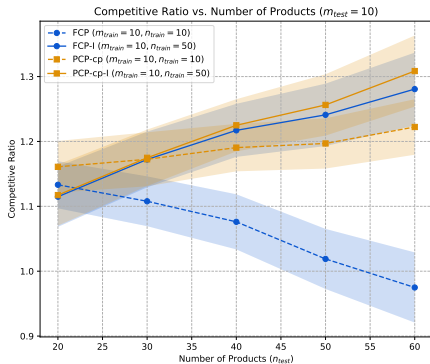
- ▶ FCP is extremely fast, but zero-shot quality declines when n becomes much larger than the training scale.
- ▶ PCP-cp retains richer bundle families and improves profit, with higher runtime.

Main read

The frontier is not solved by a single pruning rule: aggressive pruning is fast; richer candidate families buy quality. The next slide uses near-optimal labels to shift this frontier.

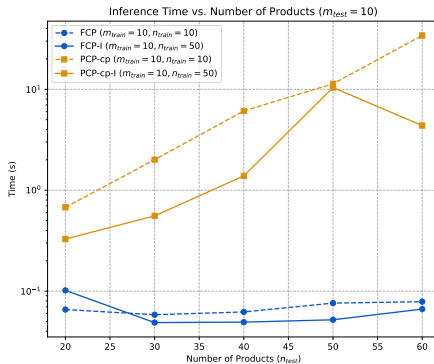
Self-improvement uses near-optimal labels to scale training

RESULTS III: ITERATIVE SELF-IMPROVEMENT



Procedure

Train on exact small labels, run PCP-cp on larger instances, then retrain the GNN on the resulting near-optimal labels at $n_{\text{train}} = 50$.



Result

Self-improvement improves transfer to $n = 60-100$, reaching about $1.28-1.33 \times$ BSP depending on the pruning policy.

Takeaways

CLOSING

1. Mixed bundling is hard because menu design and self-selection are coupled over 2^n bundles.
2. The central representation is not the bundle index; it is the segment-product projection Q^* .
3. Edge-output GNNs provide probabilities over $m \times n$ edges.
4. FCP, PCP-cp, and FCP-Local Search convert these probabilities into optimization-ready candidate families.
5. Final pricing still comes from $HM(\mathfrak{B})$, preserving the economic structure.

One-sentence takeaway

Learning identifies a compact, high-quality search space; optimization performs the final mixed-bundling pricing and self-selection.

From theory

The compact edge-output representation is expressive enough on finite solvable unfoldable datasets.

From experiments

The framework is near-optimal when exact MB is available and scalable when it is not.

Backup: full restricted MB formulation

$$\max \sum_{k=1}^m \sum_{b \in \mathcal{I}_{\mathfrak{B}}} \alpha_k Z_{kb}$$

$$\sum_{b \in \mathcal{I}_{\mathfrak{B}}} \theta_{kb} = 1, \quad \forall k$$

$$s_k \geq R_{kb} - p_b, \quad \forall k, b \in \mathcal{I}_{\mathfrak{B}}$$

$$p_b - R_{\max}(1 - \theta_{kb}) \leq P_{kb}, \quad \forall k, b \in \mathcal{I}_{\mathfrak{B}}$$

$$P_{kb} \leq p_b, \quad \forall k, b \in \mathcal{I}_{\mathfrak{B}}$$

$$s_k = \sum_{b \in \mathcal{I}_{\mathfrak{B}}} (R_{kb} \theta_{kb} - P_{kb}), \quad \forall k$$

$$R_{kb} \theta_{kb} - P_{kb} \geq 0, \quad \forall k, b \in \mathcal{I}_{\mathfrak{B}}$$

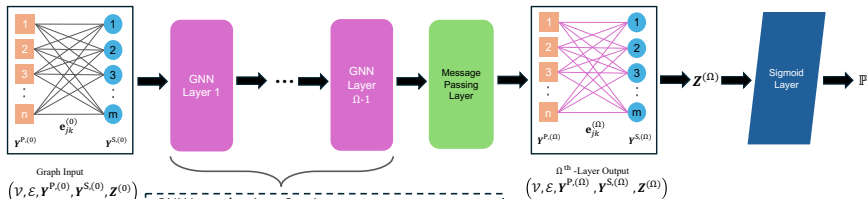
$$s_k \geq \sum_{b \in \mathcal{I}_{\mathfrak{B}}} (R_{kb} \theta_{\gamma b} - P_{\gamma b}), \quad \forall k, \gamma$$

$$Z_{kb} = P_{kb} - c_{kb} \theta_{kb}, \quad \forall k, b \in \mathcal{I}_{\mathfrak{B}}$$

$$p_b \leq \sum_{c \in \mathcal{C}} p_c, \quad \forall b, \forall \mathcal{C} \text{ covering } A(b).$$

Backup: GNN block details

ARCHITECTURE



The diagram also shows a detailed view of a **GNN Layer $l = 1, \dots, \Omega-1$** (dashed box). It consists of a **Message Passing Layer** (green rounded rectangle) followed by a **Dropout Layer** (blue parallelogram). The input to the Message Passing Layer is the **l^{th} -Layer Output**, which is a bipartite graph with nodes $1, 2, 3, \dots, n$ (orange) and $1, 2, 3, \dots, m$ (blue). The input features are $\mathbf{y}^{P,(l)}$ and $\mathbf{y}^{S,(l)}$, and the edge features are $\mathbf{e}_{jk}^{(l)}$. The output of the Message Passing Layer is the **l^{th} -Layer Output**, which is a bipartite graph with nodes $1, 2, 3, \dots, n$ (orange) and $1, 2, 3, \dots, m$ (blue). The output features are $\mathbf{y}^{P,(l)}$ and $\mathbf{y}^{S,(l)}$, and the edge features are $\mathbf{e}_{jk}^{(l)}$. The overall input to the GNN layers is $(\mathcal{V}, \mathcal{E}, \mathbf{y}^{P,(0)}, \mathbf{y}^{S,(0)}, \mathbf{Z}^{(0)})$, and the output of the Ω^{th} -layer is $(\mathcal{V}, \mathcal{E}, \mathbf{y}^{P,(\Omega)}, \mathbf{y}^{S,(\Omega)}, \mathbf{Z}^{(\Omega)})$.

$$\mathbf{m}_{\mathbf{z} \rightarrow \mathbf{w}}^{(l)} = \text{ReLU}(\mathbf{v}_{\mathbf{z}}^{(l-1)} + W_e^{(l)} \mathbf{e}_{wz}^{(l-1)}) + \epsilon \mathbf{1}, \quad \mathbf{v}_w^{(l)} = \text{MLP}_{\text{node}}^{(l)}(\mathbf{v}_w^{(l-1)} + \hat{\mathbf{m}}_w^{(l)}).$$

$$\mathbf{e}_{jk}^{(l)} = \text{LayerNorm}\left(W_2 \text{ReLU}(W_1[\mathbf{v}_j^{(l)} \parallel \mathbf{v}_{n+k}^{(l)} \parallel \mathbf{e}_{jk}^{(l-1)}] + b_1) + b_2\right).$$

Backup: pruning changes the optimization size

Policy	Bundle space	Variables	Total constraints
MB	2^n	$O(m2^n)$	$O(m2^n + m^2 + 3^n)$
FCP	$O(m)$	$O(m^2)$	$O(m^2 + m2^m)$
PCP-cp	$O(mn)$	$O(m^2 n)$	$O(m^2 n^2 + mn(n+1)^m)$
BSP	$O(n)$	$O(mn)$	$O(mn + m^2 + n^2)$

- ▶ FCP gives the strongest compression.
- ▶ PCP-cp retains a richer prefix family and uses cutting planes for cover-subadditivity.
- ▶ BSP is compact but restricts prices to be equal across bundles of the same size.

Backup: robustness and OOD checks

ADDITIONAL EVIDENCE

- ▶ Utility-function shift: testing under $f(x) = \log(1 + x)$ and $f(x) = x^{1/3}$ preserves high optimality.
- ▶ Preference-distribution shift: Beta distributions maintain strong profit ratios, showing the learned edge features are not only memorizing uniform samples.
- ▶ Seed analysis reports stable performance across random initializations.